Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Stout, Stephen C Ph.D.		
Project Title:	RNAi Analysis of Gravity Signal Transduction in Ceratopteris richardii		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stephenstout@mail.utexas.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	512-899-3130
Organization Name:	University of Texas - Austin		
PI Address 1:	Molecular Cell and Developmental Biology		
PI Address 2:	1 University Station A6700		
PI Web Page:			
City:	Austin	**State:**	TX
Zip Code:	78712	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	08/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Gravity directs polar germination in the spore of the fern Ceratopteris richardii. We are investigating the molecular basis of gravitropism in this system, using the technique of RNA interference. This technique has been used widely in other systems to suppress gene expression by the simple introduction of homologous double-strand RNA (dsRNA). By characterizing the function of these genes we will have a better understanding of gravitropism in plants.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	This research contributes to a basic understanding of plant growth and development. Although gravity is a pervasive, constant environmental signal to all terrestrial plants, plant biologists know surprisingly little about how plants sense and respond to gravity. Progress in this area will lead to a better understanding of the growth and development of plants.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)